

The Illusion of Uniqueness: Iris Recognition in Twin Subjects

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Abstract—This project evaluates iris recognition in twin subjects through a complete biometric pipeline including quality analysis, segmentation, normalization, masking, feature extraction, matching, verification, identification, and failure analysis. The system uses Daugman-style segmentation, Rubber Sheet normalization, Gabor-style binary features, and masked Hamming Distance with angular shift compensation. The experiments compare genuine pairs with different impostor classes, including unrelated subjects and twin subjects. Results show that iris texture remains discriminative in a twin-specific scenario, while twin comparisons represent a more challenging impostor condition than unrelated comparisons. Additional LBP and Blob/LoG analyses further support the presence of twin-related texture similarity.

I. INTRODUCTION

Iris recognition is one of the most widely studied biometric techniques for personal identification. The iris is particularly suitable for biometric recognition because it contains a rich texture pattern and is relatively stable over time. Unlike other biometric traits, such as face or voice, the iris is less affected by age, expression, or external appearance changes. For this reason, iris recognition systems are commonly based on the assumption that the detailed iris texture provides highly discriminative information for distinguishing individuals.

A classical iris recognition pipeline is usually composed of three main stages: iris preprocessing, feature extraction, and feature matching.

Although iris recognition is generally considered highly reliable, the case of twins is especially interesting. Twins, and in particular monozygotic twins, share the same or nearly the same genetic information and often present very similar physical characteristics. This creates a relevant question for biometric systems: if two individuals are genetically very close, does their iris texture become more similar, and can this similarity affect recognition performance?

Previous studies provide different perspectives on this question. Hollingsworth et al. observed that classical iris codes based on fine epigenetic texture details do not necessarily detect a strong similarity between genetically identical irises. However, the same work also showed that human observers can perceive some visual similarity in genetically identical irises, including the irises of identical twins. This suggests that the iris may contain different levels of information: some details are useful for biometric discrimination, while other visual patterns may reflect a more general similarity that standard biometric algorithms do not explicitly use.

A different perspective is provided by Juniati et al., who studied the similarity of iris images between twins using box-counting fractal dimension. Their work reported a significant

correlation between the iris features of twins and showed that increasing the proportion of twin iris images in the dataset reduced the recognition accuracy. This result supports the idea that twins may represent a more difficult condition for iris recognition, depending on the type of features and matching strategy used.

These observations motivate the present project. The objective is not only to build an iris recognition pipeline, but also to evaluate how the system behaves when the dataset contains twin subjects. In particular, the project investigates whether twin comparisons produce lower distances than unrelated comparisons, whether same-eye twin comparisons are more challenging than cross-eye comparisons, and whether twins are a major source of errors in verification and closed-set identification.

Code is available at https://github.com/samsungalaxys3neo/Casia_Iris_Twins.git

II. SYSTEM ARCHITECTURE AND DESIGN CHOICES

A. Architecture Overview

The application was designed as a modular biometric pipeline. Each stage receives the output of the previous one and produces intermediate files that can be inspected, reused, and evaluated independently. This structure was chosen to make the system easier to debug and to clearly separate preprocessing, feature extraction, matching, and evaluation.

The main architecture is composed of the following stages:

Table I: Architecture of the proposed iris recognition pipeline.

Stage	Role in the pipeline
Dataset audit	Reads the raw dataset, checks file structure, removes duplicates, and creates clean metadata.
Quality audit	Computes image quality descriptors such as brightness, contrast, focus, dark ratio, and bright ratio.
Segmentation	Estimates pupil and iris boundaries using a Daugman-style custom procedure.
Normalization	Converts the circular iris region into a fixed-size Rubber Sheet representation.
Masking	Selects reliable iris regions and removes noisy pixels caused by eyelids, eyelashes, reflections, or boundary errors.
Feature extraction	Extracts Gabor-style binary codes and additional LBP and Blob/LoG texture descriptors.
Pair generation	Builds genuine, twin, same-subject different-eye, and unrelated comparison pairs.
Matching	Compares iris templates using masked Hamming Distance and angular shift compensation.
Evaluation	Computes verification, identification, statistical, and failure analysis results.

B. External Libraries and Developed Code

The implementation uses standard Python libraries for low-level operations, while the biometric pipeline logic was developed specifically for this project. In particular, external libraries were used for image loading, numerical computation, filtering, table manipulation, and plotting.

Table II: Separation between library functions and custom project code.

Component	Libraries	Custom Implementation
File/Metadata	pathlib, pandas	Dataset parsing, label mgmt, duplicate handling
Image Proc.	OpenCV, NumPy	Quality audit, pre-processing, mask generation
Segmentation	OpenCV	Daugman logic, strict filtering, confidence analysis
Normalization	NumPy, OpenCV	Rubber Sheet transformation
Feature Extr.	OpenCV, NumPy	Gabor coding, LBP/Blob descriptors
Matching	NumPy	Masked HD, bit filtering, shift compensation
Evaluation	pandas	Verification, ID, twin/failure analysis

C. Main Design Choices

Several design choices were necessary in order to make the application work reliably on the available dataset.

1) *Strict segmentation subset*: **The pipeline prioritizes segmentation reliability over dataset size.**

2) *Conservative masking*: Since segmentation was not perfect, the full normalized iris strip could contain eyelids, eyelashes, reflections, sclera, or inaccurate boundary regions.

The matcher uses fewer pixels, but those pixels are more likely to represent valid iris texture.

3) *Masked Hamming Distance*: The comparison between two iris templates was performed only on pixels that were valid in both masks to avoid comparing corrupted regions.

Two templates are compared only where both images contain valid iris information.

4) *Angular shift compensation*: Small rotations of the eye or head can shift the normalized iris texture along the angular direction.

To reduce this, the final score is the best masked Hamming Distance over a limited set of angular shifts.

5) *Relation-specific pair generation*: Instead of using only a generic genuine/impostor split, the project defines multiple pair classes.

Impostor pairs are separated into meaningful biometric relations instead of being treated as a single homogeneous class.

6) *Sampling of unrelated pairs*: The number of unrelated pairs was much larger than the number of genuine and twin pairs. Using all unrelated pairs would have made the analysis unnecessarily heavy and strongly imbalanced.

For this reason, all relevant twin and genuine comparisons are preserved, while unrelated comparisons are sampled to keep the experiment computationally manageable.

7) *Auxiliary texture descriptors*: The main matcher uses Gabor-style binary codes. However, LBP and Blob/LoG descriptors were also added as an auxiliary analysis.

This produces an additional evidence for texture similarity.

III. DATASET AND PAIR DEFINITION

A. Dataset Structure

The experiments were conducted on the CASIA-Iris-Twins subset. The dataset is organized by family: each family contains two twin subjects, and for each subject both left and right eye images are available. The expected folder classes are therefore 1L, 1R, 2L, and 2R, where the number identifies the twin and the letter identifies the eye side.

For each image, the metadata extraction step associated the sample with a `family_id`, a `twin_id`, an eye side, a `subject_id`, and an `iris_id`. In this project, the biometric identity is the `iris_id`, since the left and right irides of the same subject are treated as different biometric traits.

Table III: Main metadata fields.

Field	Meaning
<code>family_id</code>	Twin family
<code>twin_id</code>	Twin 1 or twin 2
<code>eye</code>	Left or right eye
<code>subject_id</code>	Individual person
<code>iris_id</code>	One specific iris

B. Dataset Audit

An initial audit was performed before any biometric processing. The goal was to verify that all files were readable, that filenames were coherent with the folder structure, and that exact duplicates were removed.

The raw dataset contained 3183 images from 100 families. All images were readable and no filename inconsistency was detected. Two exact duplicates were found inside the same `iris_id`; they were removed to avoid artificially perfect genuine comparisons. The clean dataset therefore contains 3181 images.

Table IV: Initial dataset audit.

Quantity	Value
Raw images	3183
Families	100
Unreadable images	0
Filename errors	0
Duplicates removed	2
Clean images	3181

The image distribution across the four folder classes is reported in Table V. The dataset is not perfectly balanced, but all four twin-eye classes are sufficiently represented.

Table V: Clean image distribution.

Class	Images
1L	742
1R	831
2L	767
2R	843

C. Pairwise Comparison Classes

The central part of the evaluation is the definition of the comparison classes. Each image pair was assigned to one relation class using `family_id`, `subject_id`, `eye`, and `iris_id`. This makes it possible to separate genuine comparisons, standard impostor comparisons, and twin-specific impostor comparisons.

Table VI: Pairwise comparison classes.

Relation	Meaning
<code>genuine</code>	Same iris, different images.
<code>same_subj_diff_eye</code>	Same subject, left vs right eye.
<code>twin_same_eye</code>	Twin 1 vs twin 2, same eye side.
<code>twin_cross_eye</code>	Twin 1 vs twin 2, opposite eye side.
<code>unrelated</code>	Subjects from different families.

The `genuine` class is the positive class. All other classes are impostor comparisons. The two twin classes are kept separate because the main objective is to test whether twin comparisons (especially same-eye), are more similar than unrelated comparisons.

D. All-vs-All Pair Generation

Starting from the clean dataset, all unordered image pairs were generated. Given N images, the number of all-vs-all pairs is:

$$\frac{N(N-1)}{2}.$$

With $N = 3181$, this gives:

$$\frac{3181 \cdot 3180}{2} = 5,057,790$$

possible image pairs. These pairs were saved in compressed format and used as the metadata basis for the following verification experiments.

Table VII: All-vs-all pair counts on the clean dataset.

Relation	Pairs
<code>genuine</code>	12,361
<code>same_subj_diff_eye</code>	13,012
<code>twin_same_eye</code>	12,932
<code>twin_cross_eye</code>	13,024
<code>unrelated</code>	5,006,461
Total	5,057,790

As expected, unrelated comparisons dominate the all-vs-all set. This is typical in biometric evaluation: impostor attempts

are much more numerous than genuine attempts. For this reason, the complete all-vs-all generation is useful to define the pair classes, while later matching experiments can sample the unrelated class to reduce computational cost.

E. Role in the Final Evaluation

The pair definitions introduced in this section are used throughout the project. After segmentation, normalization, and feature extraction, the same relation logic is applied again to the final set of valid templates. In the final verification experiment, all genuine, twin, and same-subject different-eye pairs are retained, while unrelated pairs are sampled. Therefore, the final protocol is best described as a pairwise verification protocol inspired by all-vs-all evaluation, with sampled unrelated impostors.

This design allows the system to be evaluated not only as a standard genuine-versus-impostor classifier, but also as a twin-specific biometric system. In particular, it enables the direct comparison between `twin_same_eye` and `unrelated` score distributions, which is the main experimental question of the project.

IV. IMAGE QUALITY AUDIT

A. Motivation

Before performing iris segmentation, an image quality audit was carried out on the clean dataset. The purpose of this step was to verify that the images were usable and to extract simple quality indicators that could later help interpret segmentation, normalization, or matching failures.

B. Quality Descriptors

Each image was loaded in grayscale and analyzed using five global descriptors. These descriptors provide a compact description of illumination, contrast, sharpness, and possible saturation artifacts.

Table VIII: Quality descriptors computed for each image.

Descriptor	Meaning
Brightness	Mean gray-level intensity.
Contrast	Global intensity variation.
Focus score	Variance of the Laplacian.
Dark ratio	Fraction of very dark pixels.
Bright ratio	Fraction of very bright pixels.

The focus score is particularly useful because low values are usually associated with blurred images. The bright ratio and dark ratio are useful to detect strong reflections, saturated areas, or very dark regions that may affect segmentation and feature extraction.

C. Results

The quality audit was successfully completed on all 3181 images of the clean dataset. No unreadable image was found. The main statistics are reported in Table IX.

Table IX: Quality statistics on the clean dataset.

Metric	Median	Mean	Range
Brightness	126.53	125.62	68.66–205.11
Contrast	38.14	38.84	22.33–69.73
Focus	36.60	40.88	12.02–191.18
Dark ratio	0.0019	0.0034	0.0000–0.0369
Bright ratio	0.0001	0.0012	0.0000–0.3256

The brightness distribution is relatively stable, with a median value of 126.53 and a mean value of 125.62. The contrast values also indicate that most images contain sufficient gray-level variation. The focus score has a wider range, suggesting that some samples are more blurred or sharper than the average. The bright ratio is generally low, although a few images show larger saturated or very bright regions.

D. Outlier Inspection

After computing the descriptors, the most extreme samples were inspected visually. In particular, images with low focus score, low contrast, high bright ratio, and high dark ratio were selected and grouped into contact sheets.

The visual inspection showed that these outliers were not severe enough to justify automatic rejection. Even in the more difficult cases, the iris region was generally still visible. For this reason, all images were kept in the dataset.

E. Summary

The quality audit did not reduce the number of samples. Its role was mainly diagnostic. The extracted descriptors were kept in the metadata and later used to interpret difficult cases in the matching and identification stages.

Table X: Quality audit summary.

Quantity	Value
Analyzed images	3181
Unreadable images	0
Automatically rejected	0
Outliers inspected	Yes
Used for failure analysis	Yes

Since no sample was automatically rejected, the next step was to apply iris segmentation to the full clean dataset.

V. IRIS SEGMENTATION

A. Motivation

Iris segmentation is the first core step of the recognition pipeline. Its goal is to localize the pupil boundary and the external iris boundary, also called the limbus. This step is critical because the following stages depend directly on the quality of the detected iris region.

For this reason, the objective of this stage was not simply to maximize the number of accepted images, but to select a subset of reliable segmentations.

B. Preliminary Segmentation Attempts

Before selecting the final method, two preliminary segmentation strategies were tested. The first version was mainly diagnostic: it detected the pupil as a dark circular region and estimated the iris radius by searching for radial intensity changes. This approach accepted only 744 images out of 3181, corresponding to an OK rate of 23.39%, and was therefore not reliable enough for the complete pipeline.

A second custom version improved the automatic detection rate using contrast enhancement, dark-region thresholding, connected components, circularity constraints, Hough fallback, and radial edge search. This method marked 3110 images as successfully segmented, corresponding to an OK rate of 97.77%. However, visual inspection showed that this result was too optimistic: some high-confidence cases still had an inaccurate external iris circle, often too large, shifted, or affected by eyelids and eyelashes.

Table XI: Comparison of segmentation stages.

Method	Accepted	Rate
V1 diagnostic	744	23.39%
Custom V2	3110	97.77%
Daugman-style initial OK	2196	69.03%
Daugman-style strict	1465	46.05%

The preliminary experiments showed that a very high automatic acceptance rate was not necessarily a sign of good segmentation quality. For this reason, the final pipeline uses a more conservative Daugman-style segmentation followed by strict filtering.

C. Daugman-style Segmentation

The final segmentation method is inspired by the classical Daugman iris recognition pipeline. The implemented version is not the complete proprietary Daugman algorithm, but a simplified experimental approach based on the same general idea: searching for circular boundaries where the average intensity variation along the contour is high.

The procedure first applies preprocessing with Gaussian blur and CLAHE. The pupil is then localized using dark-region detection, connected components, and a Hough-based fallback when needed. After pupil localization, the external iris boundary is estimated by analyzing radial intensity variation around the pupil. The limbus search gives more importance to the lateral iris regions, because the upper and lower regions are frequently affected by eyelids and eyelashes.

The Daugman-style segmentation was applied to all 3181 clean images. The pupil was detected in all images, while the iris boundary was detected in 3162 images. Initially, 2196 images were marked as successful segmentations. The confidence distribution contained 2058 high-confidence cases, 959 medium-confidence cases, 145 low-confidence cases, and 19 cases with no valid confidence.

D. Strict Filtering

Although the Daugman-style method was more conservative than the custom V2 approach, visual inspection still showed

some imperfect iris boundaries. Therefore, an additional strict filtering step was applied before using the images for recognition.

The strict filter retained only high-confidence segmentations satisfying additional geometric and photometric constraints. In particular, it required plausible pupil and iris radii, an iris-to-pupil radius ratio between 2.0 and 3.1, a limited offset between pupil and iris centers, a sufficiently high iris score, and a dark pupil region. These constraints reduce the probability of passing bad segmentations to the normalization and matching stages.

After strict filtering, 1465 images were retained. This corresponds to 46.05% of the clean dataset. Although the filter removes more than half of the samples, it preserves 99 families out of 100 and provides a more reliable basis for the following stages.

Table XII: Final Daugman-style strict subset.

Quantity	Value
Accepted images	1465
Acceptance rate	46.05%
Families	99
Subjects	180
Iris identities	313
Iris IDs with ≥ 2 images	266
Folder class 1L	295
Folder class 1R	378
Folder class 2L	396
Folder class 2R	396

The following is an example of an accepted Daugman-style strict segmentation:

OK | conf=high | 12/1R/S3121R07.jpg

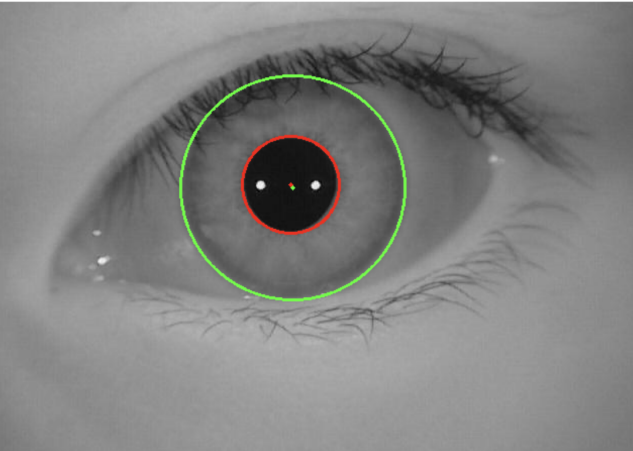


Figure 1: Example of a successful Daugman-style strict segmentation. The pupil and iris boundaries are well aligned with the visible eye structure.

The Daugman-style strict segmentation was selected as the main segmentation basis for the project. Compared with the preliminary methods, it is more conservative but more reliable.

VI. IRIS NORMALIZATION AND MASKING

A. Motivation

After segmentation, the iris region must be converted into a fixed-size representation. This step is necessary because iris images may differ in pupil size, iris radius, eye position, and small geometric variations. Even when two images belong to the same iris, their raw segmented regions are not directly comparable.

For this reason, the segmented annular iris region was normalized using a Rubber Sheet representation. The region between the pupil boundary and the outer iris boundary was unwrapped into a rectangular strip. In this representation, rows correspond to normalized radial positions and columns correspond to angular positions around the iris.

B. Rubber Sheet Representation

The normalized iris images were generated with a fixed size of 64×512 , where 64 is the radial resolution and 512 is the angular resolution. The goal of this transformation is to reduce the effect of differences in iris size, pupil dilation, and image position.

However, visual inspection showed that the full normalized strip may still contain non-iris regions. This can happen because segmentation is not perfect and because eyelids, eyelashes, reflections, and sclera may still appear inside the unwrapped iris region. Therefore, normalization alone is not sufficient: a mask is also needed to define which pixels are valid for feature extraction and matching.

C. Conservative v3 Mask

A conservative mask, called v3, was selected as the main masking strategy. The idea was to use less iris texture, but to keep only the most reliable regions. The v3 mask keeps mainly lateral iris regions, where eyelid and eyelash occlusions are usually less severe. It also excludes the most problematic radial areas, especially pixels close to the pupil and close to the external iris boundary. In addition, very bright pixels, very dark pixels, and suspicious high-gradient regions are removed.

Table XIII: Summary of the v3 normalization and masking step.

Quantity	Value
Input images	1465
Successfully normalized	1465
Failed images	0
Radial resolution	64
Angular resolution	512
Side ROI width	$\pm 38^\circ$
Valid radial rows	8–46
Mean total valid ratio	0.2365
Mean ROI valid ratio	0.9116

The total valid ratio shows that the v3 mask retains approximately 23.7% of the full normalized strip. This value is intentionally low: the objective is not to maximize the number of pixels, but to avoid comparing texture that may belong to eyelids, eyelashes, reflections, or segmentation artifacts. At the

same time, the mean ROI valid ratio is 0.9116, meaning that most pixels inside the selected lateral regions are valid.

D. Visual Inspection

The raw normalized strips may still show eyelashes or eyelid regions because they represent the complete unwrapped iris area. For this reason, the masked preview is more informative than the raw normalized image: it shows which regions are actually propagated to feature extraction and matching.

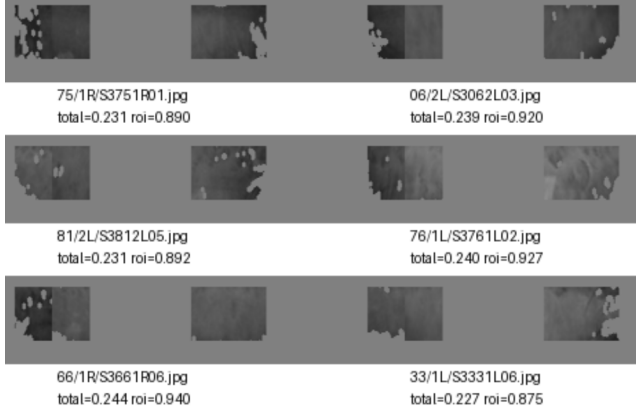


Figure 2: Examples of normalized iris strips after applying the conservative v3 mask. Gray regions correspond to pixels excluded from the following stages.

The binary mask confirms that only selected iris regions are used. In practice, the matcher does not compare the full segmented iris, but only the valid pixels preserved by the mask. This is how the pipeline reduces the effect of imperfect segmentation.

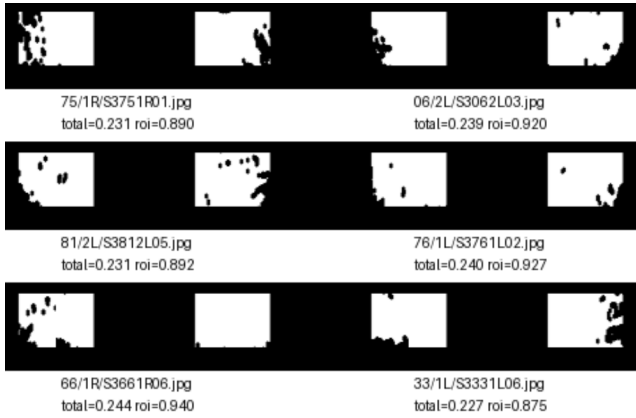


Figure 3: Examples of binary masks generated during v3 normalization. Valid regions are mainly concentrated in the lateral iris areas.

E. Role in the Pipeline

This stage is important because it compensates for one of the main limitations of the segmentation step. Since the segmentation is not perfect in every image, the pipeline does not rely on the entire segmented iris region. Instead, the

Rubber Sheet model provides a common coordinate system, while the conservative v3 mask selects only the most reliable pixels.

The output of this stage is a file containing the paths to the normalized images, the corresponding masks, and the valid ratio values. It is used as input for the following Gabor-style feature extraction step.

VII. FEATURE EXTRACTION

A. Motivation

After normalization and masking, each iris image is represented as a fixed-size rectangular strip together with a binary validity mask. The next step is to convert the normalized iris texture into a compact biometric template that can be compared efficiently.

In this project, feature extraction was performed using a Gabor-style binary coding approach. The method is inspired by the classical idea of encoding iris texture through filter responses, but it is not an implementation of the proprietary IrisCode algorithm. The objective is to obtain a binary representation that captures local iris texture while preserving the validity mask produced during normalization.

B. Gabor-style Binary Coding

For each normalized image, only the valid pixels defined by the v3 mask were used to estimate the intensity normalization parameters. This avoids letting masked regions influence the contrast normalization of the iris texture.

A bank of four Gabor-style filters was applied to each normalized iris strip. The filters combine two orientations, 0° and 90° , and two wavelengths, 8 and 16 pixels. The filter responses were binarized according to their sign:

$$b(x, y) = \begin{cases} 1, & \text{if } R(x, y) > 0, \\ 0, & \text{otherwise.} \end{cases}$$

where $R(x, y)$ is the filter response at position (x, y) . The final template contains four binary maps, one for each filter. The validity mask was propagated to the binary code, so that invalid regions are ignored during matching.

C. Feature Extraction Results

Feature extraction was successful for all normalized images. No template creation failure occurred.

Table XIV: Summary of the feature extraction stage.

Quantity	Value
Input images	1465
Feature extraction OK	1465
Feature extraction failed	0
Number of filters	4
Mean valid bit ratio	0.2151
Median valid bit ratio	0.2169
Mean bit-one ratio	0.5079

The valid bit ratio is lower than the valid pixel ratio of the normalization stage because the mask is further processed

before filtering. This is consistent with the conservative design of the pipeline: the system prefers using fewer but cleaner bits rather than including unreliable texture.

The bit-one ratio is close to 0.5. This is a positive result because it means that the binary codes are not strongly biased toward either 0 or 1. A balanced binary representation is useful for Hamming-distance-based matching, since nearly constant codes would have low discriminative power.

D. Visual Inspection

The generated binary codes were also inspected visually. The code previews show the binary response pattern produced by the filters on the normalized iris texture. These previews are not used directly for evaluation, but they are useful to verify that the feature extraction step produces structured binary patterns rather than empty or constant templates.

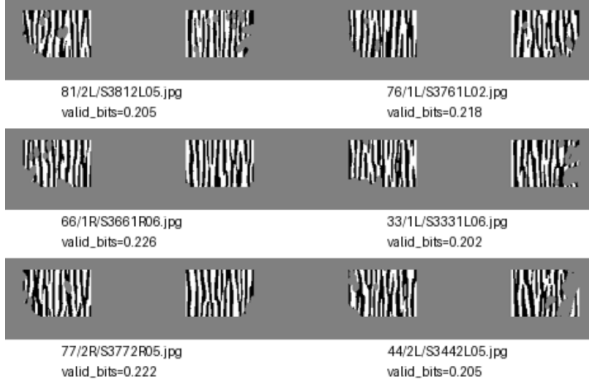


Figure 4: Examples of Gabor-style binary code previews generated from normalized and masked iris images.

The feature report is used as input for pair generation and matching.

VIII. PAIR GENERATION AND MATCHING PROTOCOL

A. Motivation

After feature extraction, each accepted iris image is represented by a Gabor-style binary code and a corresponding validity mask. The next step is to compare these templates in order to obtain biometric similarity scores.

The comparison protocol follows the pair definitions introduced in the dataset section. The goal is to evaluate not only genuine versus impostor separation, but also the specific behavior of twin comparisons.

B. Final Pair Generation

Pairs were generated using only the 1465 images for which feature extraction was successful. The total number of possible unordered pairs was:

$$\frac{1465 \cdot 1464}{2} = 1,072,380.$$

All non-unrelated pairs were retained. This includes all genuine, twin, and same-subject different-eye comparisons. Since unrelated pairs were much more numerous, they were

Table XV: Final pair generation summary.

Relation	Possible	Selected
genuine	3,984	3,984
same_subj_diff_eye	3,295	3,295
twin_same_eye	3,102	3,102
twin_cross_eye	3,008	3,008
unrelated	1,058,991	50,000
Total	1,072,380	63,389

randomly sampled to 50,000 comparisons in order to reduce computational cost while preserving a large impostor baseline.

This protocol can be interpreted as a pairwise verification protocol inspired by all-vs-all evaluation. The complete set of possible pairs was generated, but only a sampled subset of unrelated impostors was matched. This choice keeps the experiment computationally feasible while retaining all comparisons that are most relevant for the twin analysis.

C. Masked Hamming Distance

Matching was performed using masked Hamming Distance. Given two binary iris codes, the distance is computed only on bits that are valid in both masks.

The distance is defined as:

$$HD = \frac{\|(code_A \oplus code_B) \cap mask_A \cap mask_B\|}{\|mask_A \cap mask_B\|}.$$

A lower Hamming Distance indicates higher similarity. Therefore, genuine pairs are expected to have lower distances, while unrelated pairs are expected to have higher distances.

D. Angular Shift Compensation

Small rotations of the eye or small alignment errors can shift the normalized iris texture along the angular direction. To reduce this effect, angular shift compensation was applied during matching.

For each pair, the Hamming Distance was computed for several horizontal shifts:

$$s \in \{-16, -14, \dots, 14, 16\}.$$

The final score assigned to the pair was the minimum Hamming Distance obtained across all tested shifts. This follows the same general idea used in classical iris recognition systems: the matcher searches for the best angular alignment between two iris codes before making a decision.

E. Matching Results Summary

All selected pairs were successfully matched. A minimum threshold of 1000 common valid bits was required, but every comparison had a much larger number of common bits. This confirms that the conservative mask still preserves enough information for matching.

Table XVI: Matching protocol summary.

Quantity	Value
Input pairs	63,389
Successful matches	63,389
Failed matches	0
Minimum common bits	15,632
Median common bits	24,168
Mean common bits	24,136.43
Maximum common bits	30,276

IX. VERIFICATION RESULTS

A. Verification Protocol

The first evaluation scenario is biometric verification. In this setting, the system receives two iris templates and must decide whether they belong to the same iris identity. The comparison score is the masked Hamming Distance computed between two Gabor-style binary codes.

Since lower Hamming Distance indicates higher similarity, the decision rule is:

$$\text{accept if } HD \leq t,$$

where t is the decision threshold. Genuine comparisons are treated as positive attempts, while all other relation classes are treated as impostor attempts.

The verification analysis was performed separately for different impostor sets.

B. Hamming Distance Distributions

Table XVII reports the Hamming Distance statistics for each relation class.

Table XVII: Hamming Distance statistics by relation.

Relation	N	Mean	Med.	Std
genuine	3984	0.3098	0.3088	0.0691
same_subj_diff_eye	3295	0.4301	0.4290	0.0261
twin_same_eye	3102	0.4276	0.4276	0.0250
twin_cross_eye	3008	0.4331	0.4344	0.0248
unrelated	50000	0.4492	0.4490	0.0242

The average distances follow the expected trend: genuine comparisons have the lowest values, unrelated comparisons have the highest values, and twin comparisons lie between these two extremes.

This behavior is coherent with the expected biometric interpretation. Twin comparisons are more similar than unrelated comparisons, especially in the same-eye condition, but they remain clearly separated from genuine comparisons.

C. Distribution Visualization

The score distributions provide a visual interpretation of the verification problem. Genuine scores are shifted toward lower Hamming Distances, while impostor scores are shifted toward higher distances. Some overlap is still present, which explains why the system is not error-free.

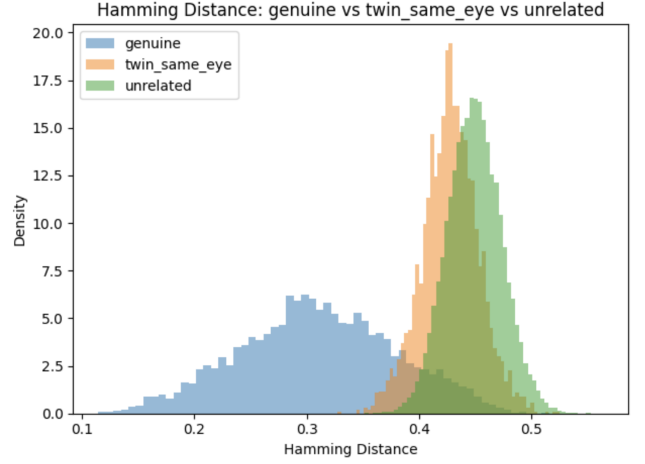


Figure 5: Hamming Distance distributions for genuine, twin same-eye, and unrelated comparisons. Genuine scores are clearly shifted toward lower distances.

The boxplot in Fig. 6 shows the same behavior from another perspective. The genuine distribution has a much lower median, while twin and same-subject different-eye comparisons are closer to the impostor region.

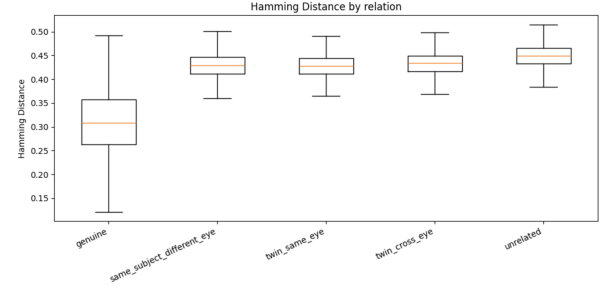


Figure 6: Boxplot of Hamming Distances by relation class. Twin comparisons show slightly lower distances than unrelated comparisons, but they remain far from genuine scores.

D. Verification Metrics

For each threshold, the following outcomes can be defined: genuine acceptances, false rejections, false acceptances, and genuine rejections. From these outcomes, the standard verification metrics are computed:

$$FAR(t) = \frac{FA}{\text{impostor attempts}},$$

$$FRR(t) = \frac{FR}{\text{genuine attempts}}.$$

The Genuine Acceptance Rate and Genuine Rejection Rate are then:

$$GAR(t) = 1 - FRR(t),$$

$$GRR(t) = 1 - FAR(t).$$

The Equal Error Rate is the operating point where FAR and FRR are approximately equal. In addition, the ROC curve summarizes the trade-off between GAR and FAR , while the DET curve shows the relationship between FRR and FAR .

Table XVIII: Verification performance for different impostor sets.

Impostor set	EER	Thr.	AUC
unrelated	0.0755	0.415	0.9735
all_impostor	0.0817	0.411	0.9693
twin_same_eye	0.1139	0.398	0.9425
twin_cross_eye	0.1041	0.402	0.9518
same_subj_diff_eye	0.1067	0.399	0.9459

The best verification performance is obtained against unrelated impostors, with an EER of 7.55% and an AUC of 0.9735. When twin same-eye comparisons are used as impostors, the task becomes more difficult: the EER increases to 11.39% and the AUC decreases to 0.9425.

This result supports the hypothesis that twin comparisons represent a more challenging impostor condition than unrelated comparisons. However, the genuine distribution remains clearly separated from the twin distributions, so the result should not be interpreted as twins being generally confused with genuine matches.

E. ROC and DET Curves

The ROC and DET curves provide a threshold-independent view of verification performance. The ROC curve shows the trade-off between genuine acceptance and false acceptance, while the DET curve highlights the balance between false rejections and false acceptances.

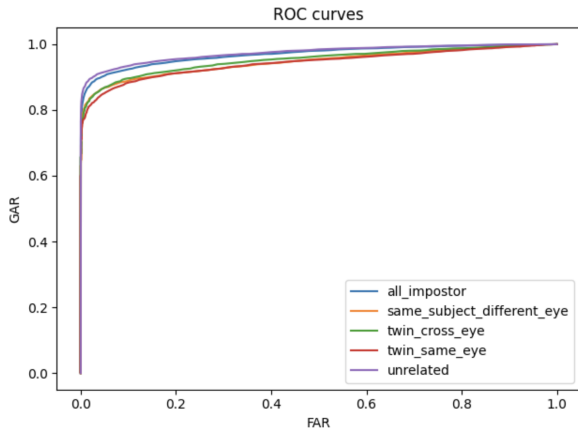


Figure 7: ROC curves for different impostor sets. The unrelated impostor condition obtains the best verification performance.

F. Interpretation

The verification experiment confirms that the proposed pipeline extracts discriminative iris information. Genuine comparisons have substantially lower Hamming Distances than impostor comparisons, and the system achieves a clear separation against unrelated subjects.

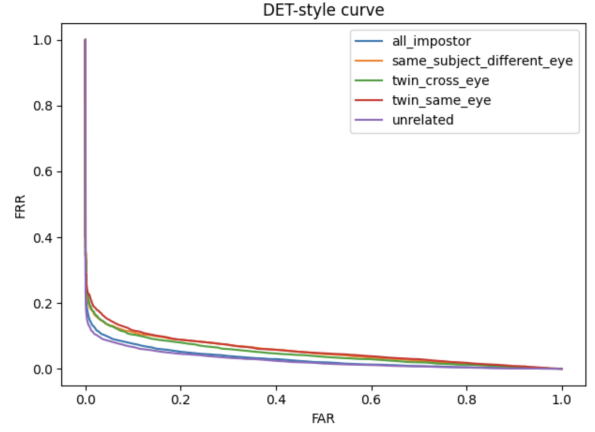


Figure 8: DET curves for different impostor sets. Twin impostors produce a more challenging error trade-off than unrelated impostors.

At the same time, the impostor class is not homogeneous. Twin same-eye comparisons produce lower distances than unrelated comparisons and lead to a higher EER. This suggests that, in the current experimental setup, twins introduce a measurable similarity effect. However, the effect is much smaller than the separation between genuine and impostor distributions.

Therefore, the main conclusion of the verification analysis is twofold. First, the system is able to distinguish same-iris comparisons from different-iris comparisons better than chance. Second, twin impostors are more challenging than unrelated impostors, especially when the same eye side is compared across twins.

X. STATISTICAL ANALYSIS OF TWIN SIMILARITY

A. Motivation

The verification results showed that twin comparisons tend to have lower Hamming Distances than unrelated comparisons. However, comparing only the mean values is not sufficient to support this observation. For this reason, a statistical analysis was performed to evaluate whether the observed differences between twin and unrelated score distributions were stable.

The main objective of this section is to test whether twin impostor comparisons, especially same-eye twin comparisons, are more similar than unrelated impostor comparisons.

B. Statistical Protocol

The analysis was performed on the Hamming Distance scores obtained from the pairwise matching experiment. The most relevant comparisons were:

- genuine vs unrelated;
- twin comparisons vs unrelated comparisons;
- twin same-eye vs unrelated same-eye;
- twin cross-eye vs unrelated cross-eye;
- twin same-eye vs twin cross-eye.

Several statistical tools were used. The Kolmogorov-Smirnov and Mann-Whitney tests were computed to compare

score distributions. In addition, bootstrap confidence intervals and permutation tests were used to estimate the stability of the mean differences.

The bootstrap procedure used 10,000 iterations to estimate the 95% confidence interval of the mean difference. The permutation test also used 10,000 iterations and tested whether the observed difference could be obtained by randomly reassigning group labels.

C. Bootstrap and Permutation Results

Table XIX reports the main comparisons. The mean difference is computed as the mean Hamming Distance of the first group minus the mean Hamming Distance of the second group. Therefore, a negative value means that the first group has lower distances and is more similar.

Table XIX: Bootstrap and permutation results.

Comparison	Diff.	Cohen's d
Genuine vs unrelated	-0.1395	-4.6587
Twin all vs unrelated	-0.0189	-0.7770
Twin same-eye vs unrelated same-eye	-0.0212	-0.8722
Twin cross-eye vs unrelated cross-eye	-0.0165	-0.6797
Same subject diff-eye vs unrelated	-0.0191	-0.7836
Twin same-eye vs twin cross-eye	-0.0055	-0.2218

For all comparisons, the 95% bootstrap confidence interval was entirely below zero. This means that the first group in each comparison consistently produced lower Hamming Distances than the second group. The permutation p-values were approximately 0.0001 in all cases, corresponding to the smallest value obtainable with the number of permutations used.

The strongest separation is observed between genuine and unrelated comparisons. This confirms that the recognition pipeline captures discriminative iris information. More importantly for the objective of the project, twin comparisons also show lower distances than unrelated comparisons.

D. Same-eye Twin Effect

The most important comparison is twin same-eye versus unrelated same-eye. This comparison is cleaner than a generic twin-versus-unrelated comparison because it controls for eye side. The observed mean difference was -0.0212, with Cohen's $d = -0.8722$.

This result suggests that, in the current experimental setup, the same eye side of two twins produces more similar iris codes than the same eye side of unrelated subjects. The effect is not as large as the genuine-versus-unrelated separation, but it is measurable and consistent.

The following figure reports the bootstrap distribution for this comparison:

E. Interpretation and Limitations

The statistical analysis supports the observation that twin iris comparisons are more similar than unrelated comparisons. This effect is strongest in the same-eye condition. However, the result must be interpreted carefully.

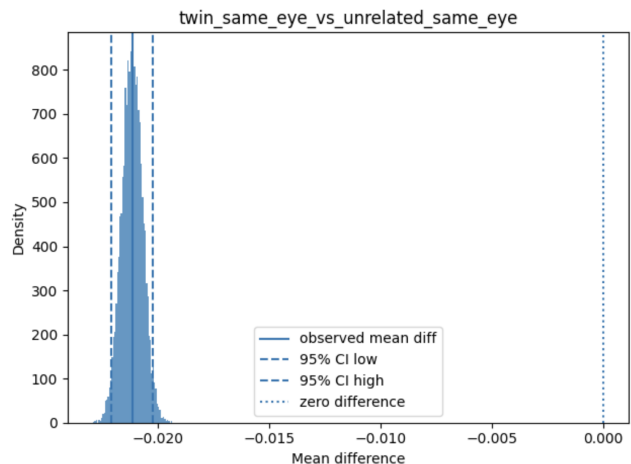


Figure 9: Bootstrap distribution of the mean difference between twin same-eye and unrelated same-eye comparisons.

First, the difference between twin and unrelated comparisons is much smaller than the difference between genuine and unrelated comparisons. Therefore, twins are not close to genuine matches; they are only slightly more similar than standard impostors.

Second, pairwise biometric comparisons are not fully independent. The same iris image can appear in multiple pairs, which may make p-values overly optimistic. For this reason, the interpretation should rely mainly on the direction of the effect, the mean differences, the bootstrap confidence intervals, and the effect sizes, rather than only on statistical significance.

XI. TEXTURE SIMILARITY ANALYSIS WITH LBP AND BLOB/LOG

A. Motivation

The main recognition pipeline is based on Gabor-style binary codes compared through masked Hamming Distance. However, to verify whether the observed twin similarity effect depends only on this representation, an additional texture analysis was performed using alternative descriptors.

Two complementary texture descriptors were extracted from the normalized and masked iris images. Local Binary Patterns (LBP) were used to describe local texture regularities through histograms computed on horizontal iris bands. In addition, Blob/LoG features were extracted using Laplacian of Gaussian responses at multiple scales, in order to capture stronger local structures of the iris texture.

This analysis is not used as the main matcher of the system. Instead, it is used as an additional experiment to check whether twin comparisons also produce lower distances than unrelated comparisons when different texture descriptors are considered.

B. Protocol

The LBP and Blob/LoG features were extracted from the same normalized images used by the main Gabor pipeline. The same pair list was then reused, so that the comparison

classes were identical to those used in the verification experiment: genuine, same-subject different-eye, twin same-eye, twin cross-eye, and unrelated.

A total of 63,389 pairs were processed, with no failed comparisons. For LBP histograms, the Chi-square distance was used. For Blob/LoG features, Euclidean distance and cosine distance were computed. In all cases, lower distance indicates higher texture similarity.

C. Results

Table XX reports the mean distance for each descriptor and relation class. The results show that genuine comparisons have the lowest distances, while unrelated comparisons have the highest distances. Twin comparisons are consistently located between genuine and unrelated comparisons.

Table XX: Mean texture distances by relation.

Metric	Gen.	S-D	T-S	T-C	Unrel.
LBP χ^2	0.4249	0.5215	0.5325	0.5458	0.6395
Blob Eucl.	0.6603	0.9078	0.9296	0.9274	1.0727
Blob Cos.	0.4538	0.7418	0.7797	0.8000	1.0025

The LBP descriptor confirms the expected ordering. Genuine comparisons have the lowest mean distance, followed by same-subject different-eye comparisons and twin comparisons. Unrelated comparisons have the highest mean distance. In particular, twin same-eye comparisons have a lower mean LBP distance than unrelated comparisons, suggesting higher texture similarity.

The Blob/LoG descriptors show the same general behavior. Both Euclidean and cosine distances are lower for twin comparisons than for unrelated comparisons. This indicates that the similarity between twins is not visible only in the Gabor-style Hamming Distance, but also in alternative texture descriptions based on local binary patterns and blob-like structures.

D. Interpretation

The additional texture analysis supports the main verification results. Twin similarity effect is not an artifact of a single matcher, but appears across different ways of describing iris texture.

At the same time, the results should be interpreted carefully. The ordering between twin same-eye and twin cross-eye comparisons is not identical for all descriptors. For LBP and Blob cosine distance, twin same-eye comparisons are slightly closer than twin cross-eye comparisons. For Blob Euclidean distance, the two twin conditions are almost equivalent, with twin cross-eye being slightly lower. Therefore, the strongest conclusion is not that same-eye twins are always the closest class, but that twin comparisons are generally more similar than unrelated comparisons.

XII. CLOSED-SET IDENTIFICATION

A. Motivation

After the pairwise verification experiment, a closed-set identification experiment was performed. In verification, the

system compares two templates and decides whether they belong to the same iris identity. In closed-set identification, instead, a probe template is compared against a gallery of enrolled identities, and the system must rank all gallery templates by similarity.

The term closed-set means that every probe is assumed to belong to one of the identities present in the gallery. Therefore, the goal is not to reject unknown users, but to determine the rank at which the correct identity appears.

B. Identification Protocol

The input of this experiment was the feature report produced after Gabor-style feature extraction.

For each `iris_id`, one image was selected as the gallery template. The selected image was the one with the highest valid bit ratio, in order to use the template with the largest amount of valid iris information. All remaining images of the same `iris_id` were used as probes.

Each probe was compared against all gallery templates using the same masked Hamming Distance and angular shift compensation used in the verification experiment. Gallery identities were then sorted by increasing Hamming Distance.

Table XXI: Closed-set identification protocol.

Quantity	Value
Input feature images	1465
Gallery templates	313
Probe templates	1152
Probe-gallery comparisons	360,576
Failed comparisons	0
Gallery selection	Best valid bit ratio

C. Identification Metrics

The main metric used in this experiment is the Rank-1 Recognition Rate, also called $\text{CMS}@1$. It measures the fraction of probes for which the correct identity is ranked first.

More generally, the Cumulative Match Score at rank k , denoted as $\text{CMS}@k$, measures the fraction of probes for which the correct identity appears within the first k candidates. The CMC curve plots $\text{CMS}@k$ as k increases.

D. Closed-set Results

The system correctly identified 952 probes out of 1152 at rank 1. This corresponds to a Rank-1 Recognition Rate of 82.64%. The cumulative match scores show that even when the correct identity is not ranked first, it often remains within the first candidates.

The median true rank was 1, confirming that most probes were correctly identified at the first position. However, the maximum true rank reached 310, showing that some probes were very difficult and were ranked close to the bottom of the gallery.

Table XXII: Closed-set identification results.

Metric	Value
Rank-1 correct	952 / 1152
Rank-1 errors	200
CMS@1	0.8264
CMS@2	0.8490
CMS@5	0.8767
CMS@10	0.8993
CMS@20	0.9158

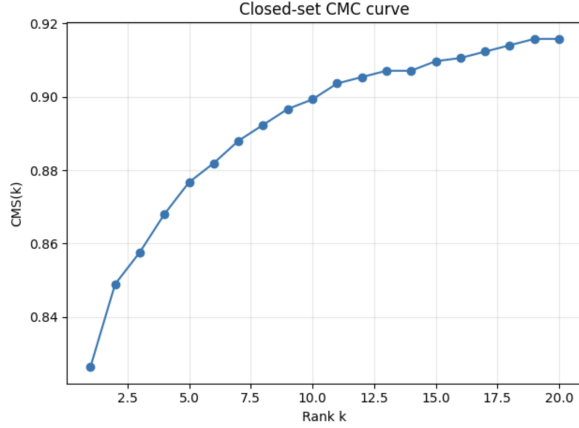


Figure 10: CMC curve for the closed-set identification experiment. The curve shows the probability that the correct identity appears within the first k candidates.

E. Identification Margin

To analyze the reliability of the identification result, the margin between the true match and the closest impostor was computed. The margin is defined as:

$$\text{margin} = \text{best impostor score} - \text{true score}.$$

A positive margin means that the true identity is closer than the best impostor. A negative margin indicates a Rank-1 identification error.

The average true match score was 0.3151, while the average best impostor score was 0.3822. The mean margin was 0.0671, with a median of 0.0640. This indicates that, in most cases, the correct gallery template was not only ranked first, but also separated from the closest impostor by a positive margin.

F. Twin-specific Analysis

A specific analysis was performed to understand whether twins were responsible for identification errors. This is important because the verification experiment showed that twin comparisons have slightly lower Hamming Distances than unrelated comparisons.

In the closed-set experiment, a twin identity was present in the gallery for most probes. However, the closest impostor was rarely a twin. Only 56 probes out of 1152 had a twin as the best impostor. Among the 200 Rank-1 errors, only 9 were caused by a twin being ranked first instead of the correct identity.

These results show that twins can appear among the top candidates in some cases, but they are not the dominant cause

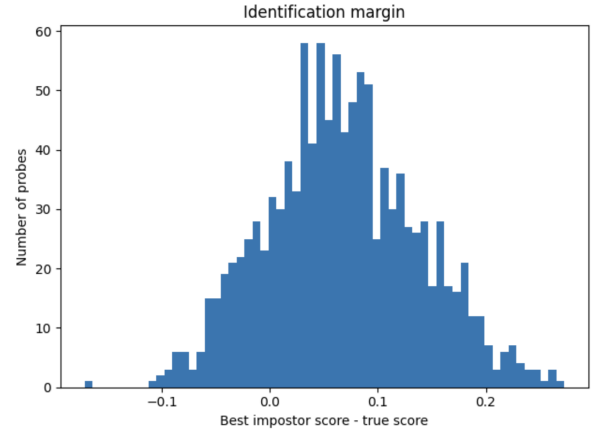


Figure 11: Distribution of the identification margin, defined as the difference between the best impostor score and the true match score.

Table XXIII: Twin analysis in closed-set identification.

Metric	Value
Probes with twin in gallery	93.32%
Best impostor is twin	4.86%
Best impostor twin same-eye	2.60%
Best impostor twin cross-eye	2.26%
Rank-1 errors caused by twin	9 / 200
Twin in top-5	13.72%
Twin in top-10	21.27%
Median best twin rank	46

of identification errors. The median rank of the closest twin was 46, meaning that in most probes the twin was not among the closest gallery identities.

G. Interpretation

The closed-set experiment confirms that the system can identify most probes correctly. The Rank-1 Recognition Rate is 82.64%, and the correct identity appears within the top 20 candidates in 91.58% of the cases.

The twin-specific analysis adds an important interpretation. In pairwise verification, twins are more similar than unrelated subjects. However, in closed-set identification, twins rarely become the closest false identity. Only 9 out of 200 Rank-1 errors were caused by twins. Therefore, in this pipeline, twin similarity increases ambiguity in some cases, but most identification errors are more likely caused by template quality, segmentation limitations, masking, or simplified feature extraction.

H. Summary

The closed-set identification experiment evaluated the system in a 1:N scenario. One template per iris identity was enrolled in the gallery, and the remaining images were used as probes. The system achieved a Rank-1 Recognition Rate of 82.64% and CMS@20 of 91.58%. Although twins were present in the gallery for most probes, they were rarely the closest impostor and caused only a small fraction of Rank-1 errors.

XIII. FAILURE ANALYSIS

A. Motivation

After the verification and closed-set identification experiments, a failure analysis was performed to understand why the system makes errors. Global metrics such as EER, AUC, and Rank-1 Recognition Rate describe the overall performance, but they do not explain which samples or comparisons are responsible for the failures.

The goal of this section is to determine whether the main errors are caused by twin similarity or by other factors, such as segmentation inaccuracies, masking limitations, low valid bit ratio, acquisition artifacts, or the simplified Gabor-style feature representation.

B. Analyzed Failure Cases

The analysis focused on difficult or suspicious cases extracted from the matching and identification results. In the pairwise verification setting, the most relevant cases are hard genuine comparisons and low-distance impostor comparisons. In the closed-set identification setting, the most relevant cases are Rank-1 errors and twin near-misses.

Table XXIV: Failure analysis input data.

Item	Count
Feature templates	1465
Pairwise matching rows	63,389
Closed-set probes	1152
Rank-1 errors	200
Twin near-miss cases	269

The hard genuine pairs are genuine comparisons with unusually high Hamming Distance. These cases indicate that two images of the same iris were represented very differently by the pipeline. Possible causes include imperfect segmentation, masking differences, occlusion, blur, or insufficient valid texture.

Low-distance impostor pairs are also important. They correspond to different identities that produced unexpectedly low Hamming Distances. These cases may be caused by true similarity, but also by non-biometric factors such as similar masks, acquisition artifacts, or limitations of the binary representation.

C. Pairwise Extreme Cases

The 40 hardest genuine pairs had a mean Hamming Distance of 0.4689 and a median of 0.4659. These values are very high for genuine comparisons and are close to the unrelated impostor region. This confirms that some same-iris comparisons are difficult for the current pipeline.

The lowest-distance impostor pairs show the opposite problem. The 40 lowest unrelated pairs had a mean Hamming Distance of 0.3642. The 40 lowest twin same-eye pairs had a slightly lower mean distance of 0.3609, while the 40 lowest twin cross-eye pairs had a mean distance of 0.3650.

These results are consistent with the previous verification analysis. Twin same-eye pairs can produce lower impostor distances than unrelated pairs. However, low-distance unrelated pairs also exist, meaning that suspicious impostor similarities are not exclusive to twins.

Table XXV: Extreme pairwise cases.

Case	Mean HD	Median HD
Hard genuine	0.4689	0.4659
Low unrelated	0.3642	0.3676
Low twin same-eye	0.3609	0.3633
Low twin cross-eye	0.3650	0.3680

D. Rank-1 Error Analysis

The closed-set identification experiment produced 200 Rank-1 errors. For each error, the relation between the probe and the wrongly selected best impostor was analyzed.

Table XXVI: Source of Rank-1 identification errors.

Best impostor relation	Errors
Unrelated	190
Twin same-eye	5
Twin cross-eye	4
Same subject different-eye	1
Total	200

This is one of the most important results of the failure analysis. Although twins were shown to be more similar than unrelated subjects in pairwise verification, they caused only 9 out of 200 Rank-1 errors. Most errors were caused by unrelated identities.

Therefore, twin similarity is not the dominant source of identification failure in this experiment. The errors are more likely related to template quality, segmentation, masking, or feature extraction limitations.

E. Valid Bit Ratio and Reliability

The valid bit ratio was also analyzed because it measures how much usable iris texture remains after normalization and masking. Rank-1 error probes had a lower average valid bit ratio than the overall probe set.

Table XXVII: Valid bit ratio comparison.

Group	Mean	Median
All probes	0.2143	0.2161
Rank-1 error probes	0.2081	0.2081

This suggests that errors are associated with a smaller amount of valid iris information. The correlation analysis supports this interpretation. Higher valid bit ratio was associated with lower genuine Hamming Distance, higher identification margin, and better true rank.

Table XXVIII: Correlations with valid bit ratio.

Correlation	Value
Genuine HD vs valid bit ratio	-0.2575
Unrelated HD vs valid bit ratio	-0.0686
Margin vs probe valid bit ratio	0.3810
True rank vs probe valid bit ratio	-0.2602

A higher valid bit ratio tends to make genuine comparisons more reliable and increases the separation margin between the true match and the closest impostor. This confirms that the amount of valid iris texture is an important factor for recognition reliability.

F. Quality-based Failure Analysis

A more detailed quality-based comparison was also performed between Rank-1 correct probes and Rank-1 error probes. The goal was to understand whether the identification failures were mainly associated with generic image quality problems, such as illumination, or with the amount of usable iris texture available after masking.

Table XXIX: Quality comparison between Rank-1 correct probes and Rank-1 error probes.

Metric	Rank-1 correct mean	Rank-1 error mean
Valid bit ratio	0.2156	0.2080
Focus score	43.41	41.45
Brightness	127.37	127.31
Dark ratio	0.0029	0.0029
Bright ratio	0.0013	0.0005

The strongest difference is observed in the valid bit ratio. Rank-1 correct probes have a mean valid bit ratio of 0.2156, while Rank-1 error probes have a lower mean value of 0.2080. This confirms that errors are more associated with a reduced amount of usable iris texture than with general illumination problems. When fewer valid bits are available, the comparison between the probe and the gallery becomes less stable, and the closest impostor can become competitive with the true match.

The focus score is also slightly lower for Rank-1 error probes. Correctly identified probes have a mean focus score of 43.41, while error probes have a mean focus score of 41.45. This suggests that reduced sharpness may contribute to some failures, because blurred or less detailed iris texture can make the extracted template less reliable. However, the difference is not as strong as the difference observed for the valid bit ratio.

Brightness, dark ratio, and bright ratio do not show the same type of separation. The mean brightness is almost identical for correct and error probes, with values of 127.37 and 127.31 respectively. The dark ratio is also the same in both groups, while the bright ratio is even lower for error probes. This indicates that Rank-1 errors are not mainly caused by images being globally too dark, too bright, or badly illuminated.

The distribution in Fig. 12 visually confirms the same result. Correctly identified probes are more concentrated toward higher valid-bit ratios, while Rank-1 error probes are more frequent at lower valid-bit ratios. Therefore, the main quality-related factor behind the errors is not simple brightness or darkness, but the amount of reliable iris texture that remains available after segmentation, normalization, and masking.

This result is coherent with the previous correlation analysis. A lower valid bit ratio reduces the amount of discriminative information used by the matcher, which can reduce the identification margin and increase the probability of a Rank-1 error. In other words, the system fails more often when it has less

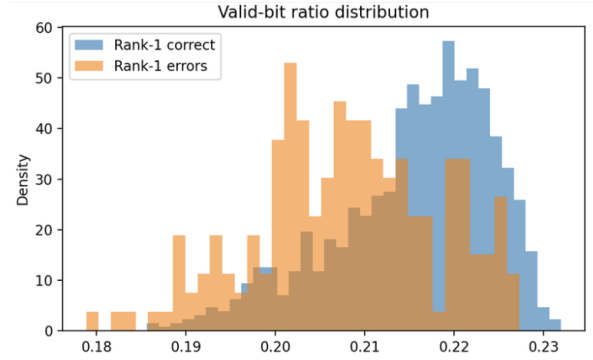


Figure 12: Valid-bit ratio distribution for Rank-1 correct probes and Rank-1 error probes. Rank-1 errors tend to be shifted toward lower valid-bit ratios, indicating that failures are associated with reduced usable iris texture.

reliable iris information to compare, not simply when the input image has different global illumination.

G. Twin Near-misses

A twin near-miss was defined as a case where the twin appeared among the top candidates or where the twin score was close to the true score. A total of 269 twin near-miss cases were identified.

In these selected cases, the closest twin had a median rank of 5. This means that when twin ambiguity occurs, the twin can appear among the first candidates. However, this result refers only to the near-miss subset. In the full closed-set experiment, the median rank of the closest twin was 46, meaning that the twin was usually not one of the closest identities.

The margin between the best twin and the true match was positive on average. Therefore, even in many near-miss cases, the true identity remained closer than the twin.

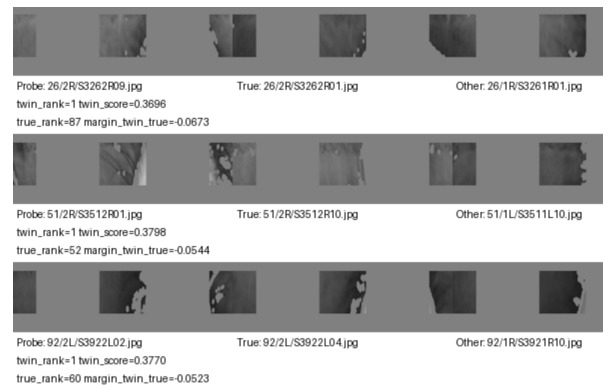


Figure 13: Examples of twin near-miss cases. These cases show situations where the twin identity appears close to the true match in the closed-set ranking.

H. Visual Failure Inspection

Visual contact sheets were generated for hard genuine pairs, low-distance impostor pairs, Rank-1 errors, and twin near-misses. These visual examples are useful to connect numerical errors with possible image-level causes.

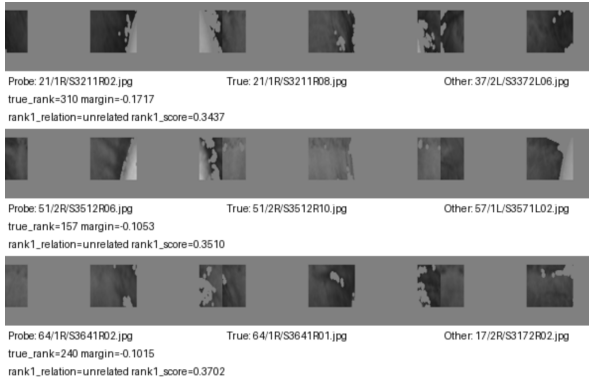


Figure 14: Examples of Rank-1 identification errors. Many errors are associated with difficult templates rather than twin identity alone.

The visual inspection supports the numerical analysis. Some failures appear related to imperfect segmentation, masked-out texture, visible artifacts, or poor template quality. This confirms that the system errors should not be interpreted only as a consequence of twin similarity.

XIV. CONCLUSION, LIMITATIONS AND FUTURE WORK

A. Conclusion

This project presented a complete iris recognition pipeline for twin subjects, including dataset organization, quality analysis, segmentation, normalization, masking, feature extraction, matching, verification, closed-set identification, and failure analysis.

The results show that the extracted iris templates contain discriminative biometric information. Genuine comparisons produced clearly lower Hamming Distances than impostor comparisons. However, twin comparisons, especially same-eye twin comparisons, were closer than unrelated comparisons, making them a more challenging impostor condition.

At the same time, twins were not systematically confused with genuine matches. In closed-set identification, twins were rarely the closest false identity and caused only a small fraction of Rank-1 errors. This suggests that twin similarity increases biometric ambiguity, but it is not the dominant source of failure.

The additional LBP and Blob/LoG texture analysis confirmed the same general trend: twin pairs were more similar than unrelated pairs also with alternative descriptors. Overall, iris recognition remains effective in a twin-specific scenario, but twin-related similarity should still be considered when evaluating biometric systems on genetically related individuals.

B. Limitations

The first limitation of the project is related to segmentation. The Daugman-style segmentation procedure used in this work is a simplified custom implementation. Although strict filtering was applied to remove the most unreliable cases, some accepted images may still contain imperfect iris boundaries,

eyelid occlusions, eyelashes, reflections, or portions of sclera. These errors can affect normalization, feature extraction, and matching.

A second limitation is the conservative masking strategy. The v3 mask was intentionally designed to keep only reliable iris regions, reducing the influence of noisy or contaminated pixels. This improves robustness, but also reduces the amount of texture available for matching. The failure analysis showed that valid bit ratio is related to recognition reliability: probes with lower valid bit ratio were more likely to produce errors or smaller identification margins.

A third limitation concerns the feature representation. The main matcher is based on Gabor-style binary codes inspired by classical iris recognition, but it is not a complete commercial or state-of-the-art iris recognition algorithm. The LBP and Blob/LoG descriptors were added as an auxiliary analysis, but they were not optimized as independent recognition systems.

Another limitation is related to the dataset. The analysis depends on the available labels, image quality, number of samples per iris identity, and balance between relation classes. In addition, pairwise comparisons are not fully independent, because the same image can appear in multiple pairs. For this reason, statistical p-values should be interpreted with caution and should not be considered the only evidence of the observed effects.

Finally, the project evaluates verification and closed-set identification, but it does not implement open-set identification. In a real biometric deployment, some probes may belong to non-enrolled identities. The system should therefore decide both whether the identity is present in the gallery and which identity it corresponds to. This would require a separate experimental protocol.

C. Future Work

A first direction for future work is to improve segmentation and masking. More advanced iris segmentation methods could better handle non-circular boundaries, eyelids, eyelashes, reflections, and off-axis acquisitions. A more accurate segmentation stage would likely increase the valid bit ratio and reduce the number of failure cases caused by poor template quality.

A second direction is to extend the project with deep learning and metric learning. Siamese Networks are particularly relevant because they are designed to compare pairs of samples. The original Siamese formulation was introduced for signature verification, where two identical subnetworks with shared weights extract features from two inputs and a distance function is used to decide whether the pair belongs to the same identity or not. This logic is very close to biometric verification, where the system must compare two biometric samples and decide whether they should be accepted as a match.

In the context of iris recognition, a Siamese architecture could replace the handcrafted Gabor-style code with a learned embedding. Two iris images would be passed through the same neural encoder, producing two feature vectors in a common embedding space. During training, genuine pairs would be pushed closer together, while impostor pairs would be pushed

farther apart. This could be done using contrastive loss or triplet loss. The resulting system could then be evaluated using the same metrics used in this project, such as EER, ROC, DET, AUC, and relation-specific score distributions.

This direction is especially interesting for twin subjects. Recent work by Yuan and Bowyer proposed a Siamese Network with contrastive learning to classify pairs of iris images as monozygotic or non-monozygotic. Their study is directly connected to the problem investigated in this project because it does not only ask whether two irises belong to the same identity, but whether two iris images contain enough information to infer genetic relatedness. In that work, the model was trained and tested using different input conditions: original eye images, iris-only images, and non-iris-only images. This is important because it suggests that both iris texture and surrounding ocular information may contribute to the distinction between monozygotic and non-monozygotic pairs.

A natural extension of this project would therefore be to train a Siamese model on the current dataset using the pair labels already defined in the pipeline. The training pairs could be organized into genuine, twin same-eye, twin cross-eye, same-subject different-eye, and unrelated comparisons. Instead of using a single binary genuine/impostor label, the model could be trained with a relation-aware strategy, giving special attention to hard impostor classes such as twin same-eye pairs.

Another future direction is the classification of zygosity. The current project distinguishes between twin and unrelated comparisons, but it does not separate monozygotic and dizygotic twins. A stronger dataset with reliable zygosity labels would make it possible to study whether biometric traits can help distinguish monozygotic twins, dizygotic twins, and unrelated individuals. This would be useful not only for iris recognition, but also for a broader biometric analysis of genetically related subjects.

Machine learning and deep learning could also be used to compare different biometric traits in the same zygosity classification problem. Possible traits include iris texture, periocular appearance, face, fingerprints, palmprints, voice, gait, and ear shape. A multimodal approach could combine several sources of evidence and evaluate whether fusion improves the separation between monozygotic twins, dizygotic twins, and unrelated individuals.

For iris specifically, future work could compare three levels of ocular information. The first level is the normalized iris texture alone. The second level is the periocular region, including eyelids, eye shape, eyelashes, and surrounding skin texture. The third level is the full eye image, combining iris and periocular cues. This comparison would help determine whether twin-related similarity is mainly located in the iris texture itself or in the surrounding ocular region.

A further extension would be to use explainable deep learning. Attention maps, Grad-CAM, saliency maps, or other attribution methods could help identify which parts of the eye region are most important when the model classifies two samples as monozygotic, dizygotic, or unrelated. This would make the system more interpretable and would help connect the learned features with biological or anatomical structures.

Finally, the project could be extended with a more complete biometric evaluation protocol. Future experiments could include open-set identification, cross-dataset testing, subject-disjoint train/test splits, stronger statistical validation, and comparison with established iris recognition baselines. These improvements would make it possible to better understand how robust the observed twin similarity effect is across different datasets, algorithms, and biometric scenarios.

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